

On the Unique Catalan Identity in Ramanujan's Nested Radical Family

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May 5, 2026

Abstract

For each integer $n \geq 1$, Ramanujan's nested radical identity

$$n + 1 = \sqrt{1 + n\sqrt{1 + (n + 1)\sqrt{1 + (n + 2)\sqrt{\cdots}}}}$$

holds, with first-level unfolding $(n + 1)^2 = 1 + n(n + 2)$. We show that the case $n = 2$, yielding $3^2 - 2^3 = 1$, is the unique member of the family for which the quantity $n(n + 2)$ is a non-trivial perfect power. The proof reduces to two elementary observations and an application of Mihailescu's theorem [3], the resolution of Catalan's conjecture. We comment on what this says about the place of the case $n = 2$ within the Ramanujan family.

1. Introduction

In a problem submitted to the *Journal of the Indian Mathematical Society* [4], Ramanujan asked the reader to evaluate

$$\sqrt{1 + 2\sqrt{1 + 3\sqrt{1 + 4\sqrt{1 + 5\sqrt{1 + \cdots}}}}}, \quad (1)$$

giving the answer 3. This nested radical is one of the most famous identities in elementary arithmetic. It is in fact a special case of the general formula recorded in Ramanujan's *Notebooks* [1, Ch. 12]:

$$n + 1 = \sqrt{1 + n\sqrt{1 + (n + 1)\sqrt{1 + (n + 2)\sqrt{\cdots}}}}, \quad n \in \mathbb{N}_{\geq 1}. \quad (2)$$

The convergence of the right-hand side as a limit of finite truncations was settled by Vijayaraghavan [5] and Herschfeld [2]; the algebraic identity itself follows from the recursive relation $f(n)^2 = 1 + n \cdot f(n+1)$ with $f(n) = n+1$, since $(n+1)^2 - 1 = n(n+2)$.

The case $n = 2$ in (2) recovers (1). Setting aside its visual elegance, it is natural to ask whether $n = 2$ enjoys any *intrinsic* singularity within the family (2). At first glance, no: the identity is parametric, and every $n \geq 1$ produces a perfectly valid nested radical. In this note we observe that, when one looks at the first-level expansion

$$(n+1)^2 = 1 + n(n+2), \quad (3)$$

the case $n = 2$ does enjoy a singular property: it is the *unique* value of n for which $n(n+2)$ is a non-trivial perfect power, and accordingly the unique value for which (3) is itself a Catalan-type identity

$$3^2 - 2^3 = 1. \quad (4)$$

The core of the observation is that this uniqueness is not merely empirical: it is a consequence of Mihailescu's theorem [3], the proof of Catalan's conjecture. The purpose of this note is to record this remark, give a clean proof, and discuss what it says about the place of $n = 2$ in the Ramanujan family.

2. The Result

Theorem 1. *For $n \in \mathbb{Z}_{\geq 1}$, the integer $n(n+2)$ is a non-trivial perfect power, i.e. there exist $b, q \in \mathbb{Z}_{\geq 2}$ with*

$$n(n+2) = b^q,$$

if and only if $n = 2$, in which case $(b, q) = (2, 3)$ and the identity (3) reduces to $3^2 - 2^3 = 1$.

Proof. We split according to the value of $q \geq 2$.

Case 1: $q = 2$. Suppose $n(n+2) = b^2$ for some $b \geq 2$. Using $n(n+2) = (n+1)^2 - 1$, this is

$$(n+1)^2 - b^2 = 1, \quad \text{i.e.,} \quad ((n+1) - b)((n+1) + b) = 1. \quad (5)$$

Since $b \geq 2$ and $n \geq 1$, both factors on the left of (5) are positive integers; the only factorisation of 1 in $\mathbb{Z}_{>0}$ is $1 \cdot 1$, forcing $(n+1) - b = (n+1) + b = 1$, hence $b = 0$ and $n+1 = 1$, contradicting $b \geq 2$ and $n \geq 1$. Thus no $n \geq 1$ makes $n(n+2)$ a perfect square.

(Equivalently: $(n+1)^2 - 1$ lies strictly between the consecutive squares n^2 and $(n+1)^2$ for $n \geq 1$, hence is not a square.)

Case 2: $q \geq 3$. Suppose $n(n+2) = b^q$ for some $b \geq 2$ and $q \geq 3$. Then

$$(n+1)^2 - b^q = 1, \quad (6)$$

which is the Catalan equation $a^p - b^q = 1$ with parameters $(a, p, q) = (n + 1, 2, q)$. By Mihailescu's theorem [3], the unique solution in positive integers to

$$a^p - b^q = 1, \quad a, b, p, q \geq 2,$$

is $(a, b, p, q) = (3, 2, 2, 3)$. Specialising to our case ($p = 2, q \geq 3$) yields $a = n + 1 = 3$, $b = 2, q = 3$, and therefore $n = 2$ with $n(n + 2) = 2 \cdot 4 = 8 = 2^3$.

Combining the two cases, $n(n + 2) = b^q$ with $b, q \geq 2$ forces $n = 2, b = 2, q = 3$, and (3) becomes the identity $3^2 - 2^3 = 1$. Conversely, $n = 2$ obviously gives such a representation. $\square$

3. Remarks

3.1. Depth of the theorem versus simplicity of the application.

Theorem 1 reduces to a one-line application of Mihailescu [3], but the underlying theorem is deep: its proof rests on the theory of cyclotomic units and primary roots in cyclotomic fields, building on earlier reductions by Cassels, Inkeri and others. What is novel here is not the depth, but the observation that this depth manifests itself in an entirely elementary context, that of Ramanujan's nested radicals. The theorem is, in a sense, a translation of Mihailescu into the Ramanujan family.

3.2. Higher levels of the radical.

The first-level expansion (3) is special. At depth k , repeated unfolding gives

$$(n + 1)^2 = 1 + n f(n + 1) = 1 + n \sqrt{1 + (n + 1) \sqrt{1 + (n + 2) \sqrt{\cdots}}},$$

and the values appearing inside no longer have the structure of $(n + 1)^2 - 1$. Whether some *deeper* level of the radical for some other n produces a Catalan-type identity is a separate question; we do not address it here, but note that it would require a non-elementary perfect-power identity in $\sqrt{1 + (n + 1) \sqrt{\cdots}}$, which is not known to occur.

3.3. Connection to Pillai's conjecture.

Pillai's conjecture asserts that for every $k \geq 1$, the equation $a^p - b^q = k$ has finitely many solutions in (a, b, p, q) with $a, b, p, q \geq 2$. The Catalan case $k = 1$ is now known (Mihailescu); higher k remain open. Our observation can be rephrased as: *within the Ramanujan family* (2), the perfect-power difference $|a^p - b^q|$ at the first level reaches the value 1 exactly once, at $n = 2$. Whether the same can be said for higher gap values

$|a^p - b^q| = k$, $k \geq 2$, depends on parametric solutions of $n(n+2) - b^q = \pm k$, where partial results (e.g. Bilu, Bugeaud) apply but no general analogue of Mihailescu's theorem is available.

3.4. Visual interpretation.

Theorem 1 can be paraphrased as follows. The family (2) realises every positive integer $n+1$ as a nested radical; among these, the integer 3 is the only one whose *innermost defining product* $(2 \cdot 4)$ collapses multiplicatively into a single perfect power (2^3) . This is what makes the case $n=2$, of all the visually similar cases in the family, arithmetically singular.

References

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